

distribution for the horizon concerned. Many of our knowledge problems come from this tension between a priori and a posteriori.

Proof in the Flesh

There is no reliable way to compute small probabilities. I argued philosophically the difficulty of computing the odds of rare events. Using almost all available economic data—and I used economic data because that’s where the clean data was—I showed the impossibility of computing *from the data* the measure of how far away from the Gaussian one was. There is a measure called kurtosis that the reader does not need to bother with, but that represents “how fat the tails are,” that is, how much rare events play a role. Well, often, with ten thousand pieces of data, forty years of daily observations, one single observation represents 90 percent of the kurtosis! Sampling error is too large for any statistical inference about how non-Gaussian something is, meaning that if you miss a single number, you miss the whole thing. The instability of the kurtosis implies that a certain class of statistical measures should be totally disallowed. This proves that everything relying on “standard deviation,” “variance,” “least square deviation,” etc., is bogus.

Further, I also showed that it is impossible to use fractals to get acceptably precise probabilities—simply because a very small change in what I called the “tail exponent” in [Chapter 16](#), coming from observation error, would make the probabilities change by a factor of 10, perhaps more.

Implication: the need to avoid exposure to small probabilities in a certain domain. We simply cannot compute them.

FALLACY OF THE SINGLE EVENT PROBABILITY

Recall from [Chapter 10](#), with the example of the behavior of life expectancy, that the conditional expectation of additional life drops as one advances in age (as you get older you are expected to live a smaller number of years; this comes from the fact that there is an asymptotic “soft” ceiling to how old a human can get). Expressing it in units of standard deviations, the conditional expectation of a Mediocristani Gaussian variable, conditional on it being higher than a threshold of 0, is .8 (standard deviations). Conditional on it being higher than a threshold of 1, it will be 1.52. Conditional on it being higher than 2, it will be 2.37. As you see, the two numbers should converge to each other as the deviations become large, so conditional on it being higher than 10